2. To test the equality of two population means when population variances are equal and unknown, using T_2.
3. To test the equality of two means – paired t-test, based on dependent samples, T_3.

YOU WILL KNOW

The t-distribution has few more applications but they are not considered in this Chapter. You will study these applications in higher classes.

2.1.4 Test of Hypotheses for Normal Population Mean (Population Variance is Unknown)

Procedure:

Step 1 : Let μ and σ^2 be respectively the mean and variance of the population under study, where σ^2 is unknown. If μ_0 is an admissible value of μ, then frame the null hypothesis as

H_0: $\mu = \mu_0$ and choose the suitable alternative hypothesis from

(i) H_1: $\mu \neq \mu_0$ (ii) H_1: $\mu > \mu_0$ (iii) H_1: $\mu < \mu_0$

Step 2 : Describe the sample/data and its descriptive measures. Let $(X_1, X_2, \ldots, X_n)$ be a random sample of n observations drawn from the population, where n is small ($n < 30$).

Step 3 : Specify the level of significance, α.

Step 4 : Consider the test statistic $T = \dfrac{\bar{X} - \mu_0}{S/\sqrt{n}}$ under H_0, where $\bar{X}$ and S are the sample mean and sample standard deviation respectively. The approximate sampling distribution of the test statistic under H_0 is the t-distribution with $(n-1)$ degrees of freedom.

Step 5 : Calculate the value of t for the given sample $(x_1, x_2, \ldots x_n)$ as $T = \dfrac{\bar{x} - \mu}{s/\sqrt{n}}$.

here $\bar{x}$ is the sample mean and $s = \sqrt{\dfrac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$ is the sample standard deviation.

Step 6 : Choose the critical value, t_e, corresponding to α and H_1 from the following table

Alternative Hypothesis (H_1)	$\mu \neq \mu_0$	$\mu > \mu_0$	$\mu < \mu_0$
Critical Value (t_e)	$t_{n-1,\alpha/2}$	$t_{n-1,\alpha}$	$-t_{n-1,\alpha}$

Step 7 : Decide on H_0 choosing the suitable rejection rule from the following table corresponding to H_1.

Alternative Hypothesis (H_1)	$\mu \neq \mu_0$	$\mu > \mu_0$	$\mu < \mu_0$
Rejection Rule	$\lvert t_0 \rvert \geq t_{n-1,\alpha/2}$	$t_0 > t_{n-1,\alpha}$	$t_0 < -t_{n-1,\alpha}$

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